

Ayuba Yusufu

✉ Johnyusufuayuba@gmail.com ☎ +44 (0) 7490524299 📍 Nottingham, UK
■ <https://olympus.ntu.ac.uk/n1354251>

Summary

Student with practical knowledge in C++, Java and Python experienced in system architecture and building user-friendly applications. Passionate about leveraging technology to tackle practical issues and transform complex concepts into practical solutions. A Strong background in problem-solving and critical thinking, with a track record of delivering scalable projects through teamwork.

Professional Experience

Hospitality Kiosk

Mint People

04/2025 – Present | Nottingham, UK

- During busy times, I efficiently and precisely handle card payments, minimizing mistakes, and upholding stadium finance procedures.
- Ensure sales and seamless operations are facilitated by monitoring kiosk stock levels, quickly restocking goods, and minimizing shortages.
- Ensure compliance with safety and food handling regulations.

Education

BSc Computer Science

Nottingham Trent University

2024 – Present | Nottingham, UK

Cloud-Based Load Balancer with Distributed File Management

COMP20081 – Systems Software | NTU | Individual Coursework (80% of module)

Java · JavaFX · Docker · SQLite · MySQL · MQTT · SSH (JSch) · AES-128 ·

PBKDF2 · Zip4j · CMake

- Designed and deployed an 8-container Docker Compose system: load balancer (FCFS/SJN/Round Robin scheduling), MQTT broker, file partitioner, dual storage nodes, LAMP server, and main application VM.
- Built a full-stack JavaFX desktop application (8 screens) with UUID session tokens, role-based ACL for file sharing, and a built-in terminal emulator.
- File chunking system: 1024-byte AES-128 CBC encrypted chunks distributed round-robin via SSH; three-layer encryption (PBKDF2/HMAC-SHA1, Zip4j AES, AES-128 CBC).
- Dual-database persistence (SQLite + MySQL) with 60-second sync, MQTT real-time messaging (5 topics, QoS 1), and SSH container health monitoring every 10 seconds.
- Performance metrics tracked (throughput, wait/turnaround, per-container stats); all 20 coursework requirements including all advanced features implemented.

Voting Analysis Programme

- Developed a Python console-based voting analysis software that can read, write, and remove voter information from CSV files
- For effective data administration, strong error handling and a user-friendly interface were implemented

Online Recruitment System

- Collaborated in a six-person team to develop an online recruitment system
- Led UI design using best practices for intuitive user experience
- Contributed to functional analysis and system architecture to guarantee usability and scalability

Contact Management System

Skills

Technical Skills

- C++
- Java
- Python
- JavaFX
- Docker
- SQLite
- MySQL
- MQTT
- SSH
- CMake

Soft Skills

- Teamwork
- Critical Thinking
- Time Management
- Problem Solving

Interests

- Basketball
- Reading and Research
- Coding

International Foundation in Science and Engineering

INTO Manchester

01/2024 – 07/2024 | Manchester, UK

Core subjects:

- Chemistry
- Mathematical Engineering
- Physics
- EAP (English for Academic Purposes)

Certificates

Ethical Hacking Fundamentals

[2022]